Notes: Schlafmann, Setty & Vestman (RFS, 2026)
Designing Pension Plans According to Consumption-Savings Theory

Contents

1 Big picture

- **Question.** How should contribution rates in mandatory defined contribution (DC) pension plans be designed if the goal is to approximate optimal consumption-savings behavior over the life cycle?
- **Core critique.** Existing mandatory DC plans commonly use a constant contribution rate as a fraction of labor income. The paper argues that this is not consistent with basic consumption-smoothing theory.
- **Two guiding principles.** Optimal contribution rates should: (i) increase with age, because young workers typically expect income growth and would like to postpone saving; and (ii) decrease with the DC account balance-to-income ratio, because high accumulated balances or temporarily low income imply less need for current saving.
- **Empirical fact.** Swedish registry data show that many people adjust liquid nonmandated saving in a way broadly consistent with these principles, but almost half the population is hand-to-mouth and cannot offset mandated constant contributions.
- **Policy proposal.** The paper designs simple, automatic DC contribution-rate rules that depend on age and the balance-to-income ratio while maintaining the same average replacement rate.
- **Quantitative result.** A combined age and balance-to-income rule produces a welfare gain of about 1.8 percent in consumption-equivalent terms and reduces dispersion in replacement rates.
- **Robustness.** The benefits survive temptation/self-control preferences, alternative risk aversion, alternative elasticity of intertemporal substitution, and comparisons to alternative investment and annuitization reforms.
- **Takeaway.** Mandatory pension design should focus not only on asset allocation or decumulation, but also on the time path and state dependence of contributions.

2 Incremental contribution of the paper

- **From contribution levels to contribution rules.** The paper shifts pension-design analysis away from choosing a single constant mandatory contribution rate and toward designing state-contingent contribution rules.
- **Consumption-savings principles in DC plan design.** The paper derives simple life-cycle principles for optimal savings rates: increasing with age and decreasing in the asset balance-to-income ratio.
- **Empirical bridge from household saving to pension design.** Using Swedish registry data, the paper asks whether households already implement these principles in liquid nonmandated saving and shows large heterogeneity by hand-to-mouth status.
- **Heterogeneity and liquidity constraint insight.** The main welfare motivation is not that all households make mistakes; it is that almost half the population has too little liquid wealth to offset rigid mandatory contributions.
- **Quantitative policy rule.** The paper translates theory into an implementable rule using only variables observable to pension administrators: age and DC account balance-to-income.
- **Replacement-rate distribution metric.** The paper evaluates reforms not only by average replacement rates, but also by dispersion, showing that flexible contributions reduce both very low and very high retirement-income replacement outcomes.
- **Robustness to behavioral motives.** The paper shows that flexible contributions can remain valuable even under temptation preferences, weakening the objection that mandatory constant contributions are needed to protect present-biased households.
- **Policy-margin comparison.** The paper demonstrates that redesigning contribution rates yields welfare effects more than an order of magnitude larger than changing DC asset allocation or annuitization margins.

Incremental contribution in one sentence. The paper turns consumption-savings theory into a practical design principle for mandatory DC pensions: contribution rates should be automatically age dependent and balance-to-income dependent, and this design yields large welfare gains especially for households that cannot undo constant mandatory contributions.

3 Data and measurement

3.1 Swedish registry data

- The empirical analysis uses representative Swedish registry data for 2000-2007.
- The data contain detailed balance-sheet, income, demographic, and portfolio information.
- The setting is useful because Sweden has large mandatory pension contributions and detailed wealth-tax-based asset reporting during the sample period.
- The data allow the authors to measure active saving in liquid financial wealth outside the pension system.

Interpretation: The empirical part asks whether people voluntarily follow consumption-savings principles when they have liquid nonmandated assets. This is informative because mandated pension contributions themselves are rigid.

3.2 Savings rate in liquid financial wealth

The paper constructs active saving from changes in financial wealth net of passive returns:

$$\Delta\widetilde{A}_{it} = A_{it} - A_{it-1}R^{A}_{it}.$$

The active savings rate is

$$\widetilde{\lambda}_{it} = \frac{\Delta\widetilde{A}_{it}}{Y^{Disp}_{it}},$$

where Y^{Disp}_{it} is disposable income.

Interpretation: This measure captures discretionary liquid saving or dissaving outside the pension system. It is the empirical counterpart to a pension contribution rate.

3.3 Hand-to-mouth status

- The paper defines hand-to-mouth households as those with liquid financial wealth less than or equal to one-twelfth of disposable income.
- About 47.2 percent of individuals are hand-to-mouth in the full sample.
- Hand-to-mouth status is much more common among nonparticipants in the stock market than among stock market participants.

Interpretation: Hand-to-mouth households cannot easily undo mandatory constant contributions by adjusting liquid saving. This is why redesigning mandatory contributions can improve welfare.

3.4 Quantitative model inputs

- The model follows individuals from age 25 to at most age 100, with retirement at 65.
- It includes Epstein-Zin preferences, risky labor income, stock and bond returns, stock market participation costs, liquid financial wealth, a notional pension account, and a funded DC account.
- The model is calibrated to match asset-to-income ratios, stock market participation, stock shares, and the distribution of financial wealth and income over the life cycle.

Interpretation: The quantitative model connects the empirical facts to policy counterfactuals. It is rich enough to evaluate contribution-rate design in a realistic Swedish pension environment.

4 Models and empirical specifications

4.1 Stylized three-period model

Individuals maximize

$$\log C_{i,1} + \beta \log C_{i,2} + \beta^2 \log C_{i,3}$$

subject to

$$C_{i,1} = Y_{i,1} - A_{i,1}, \quad C_{i,2} = A_{i,1}R_1 + Y_{i,2} - A_{i,2}, \quad C_{i,3} = A_{i,2}R_2.$$

Savings rates are

$$\lambda_{i,1} = \frac{A_{i,1}}{Y_{i,1}}, \qquad \lambda_{i,2} = \frac{A_{i,2} - A_{i,1}R_1}{Y_{i,2}}.$$

Interpretation: The model isolates the two core design variables for DC contribution rates: age/income-growth stage and accumulated balance relative to income.

4.2 Optimal savings-rate principles

The model yields:

$$\lambda_1^* = \frac{\beta + \beta^2}{1 + \beta + \beta^2} - \frac{1}{1 + \beta + \beta^2} \frac{1}{R_1} \frac{Y_{i,2}}{Y_{i,1}},$$

so young-age saving falls with expected income growth. It also yields:

$$\lambda_2^* = \frac{\beta}{1 + \beta} - \frac{1}{1 + \beta} \frac{A_{i,1}R_1}{Y_{i,2}},$$

so middle-age saving falls with the balance-to-income ratio.

Interpretation: These closed-form expressions are the conceptual heart of the paper. Constant contribution rates are optimal only in knife-edge circumstances.

4.3 Empirical tests of the principles

The paper estimates how liquid saving responds to expected income growth:

$$\widetilde{\lambda}_{it} = \beta_0 + \beta_1 (y_{i,t+s}^{Disp} - y_{it}^{Disp}) + \epsilon_{it},$$

and how saving responds to asset-balance and income shocks. It uses IV designs based on asset-type returns and employer wage growth to address endogeneity.

Interpretation: The empirical tests ask whether households behave as the stylized model predicts when they have enough liquid wealth to adjust.

4.4 Flexible contribution rule

The policy rule can be summarized as

$$\lambda_{it} = \beta_0 + \beta_1 Age_{it} + \beta_2 \left(\frac{B_{it}}{Y_{it}} / \chi_t - 1 \right),$$

with variants using only age, only balance-to-income, or both.

Interpretation: This rule is deliberately simple. It uses information observable to plan administrators and automates the two theoretical principles.

5 Identification and empirical design

- **Empirical principle tests.** Expected income growth is proxied with future income growth; asset-balance shocks are instrumented with asset-type portfolio returns; income shocks are instrumented using employer wage growth.
- **Key heterogeneity.** The paper consistently separates hand-to-mouth and non-hand-to-mouth households. This is essential because liquid-wealth constraints determine whether households can offset rigid pension contributions.
- **Model-based policy evaluation.** The welfare analysis is structural and calibrated. It does not rely on a randomized policy experiment, but on a quantitative life-cycle model disciplined by Swedish registry moments.
- **Policy counterfactuals.** The benchmark keeps constant contribution rates and current pension rules. Counterfactuals redesign contribution rates while preserving the same average replacement rate.
- **Robustness.** The paper checks temptation preferences, preference parameters, investment strategy changes, and annuitization changes.
- **Limitations.** The main policy evaluation depends on model structure and calibration. The empirical tests show household behavior in liquid assets, but the welfare gains come from structural counterfactuals.

6 Section-by-section study guide

- **Introduction.** Motivates why constant DC contribution rates are inconsistent with consumption-savings theory and states the welfare result.
- **Section 1.** Derives the two theoretical principles in a stylized three-period model: age dependence and balance-to-income dependence.
- **Section 2.** Uses Swedish registry data to test whether liquid savings behavior follows the principles and highlights hand-to-mouth heterogeneity.
- **Section 3.** Builds the quantitative life-cycle model with detailed pension accounts, stock market participation, risky labor income, and annuitization.
- **Section 4.** Designs flexible contribution rules and reports welfare, replacement-rate, behavior, and cash-flow effects.
- **Section 5.** Tests robustness to temptation preferences, risk aversion, EIS, investment strategy, and annuitization.

- **Conclusion.** Argues that contribution-rate design is a first-order policy margin for DC pension systems.

7 Tables and figures

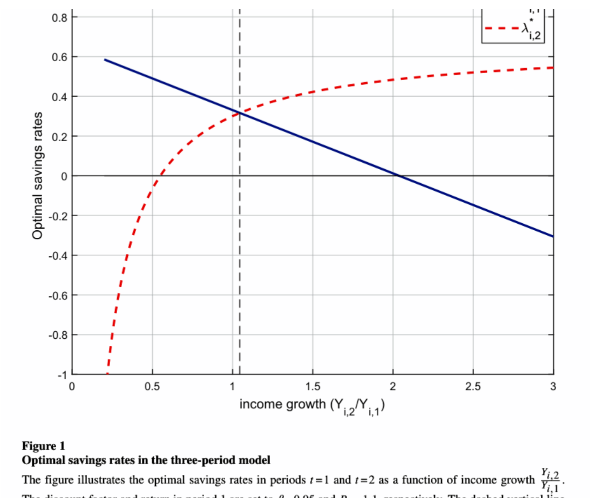
7.1 Figure 1 and Table 1: Model principle and empirical sample

Read:

- Figure 1 plots optimal savings rates in the three-period model against income growth.
- The young-period savings rate falls with income growth; the middle-period savings rate rises toward a positive rate.
- The vertical dashed line shows the knife-edge case in which constant savings rates are optimal.
- Table 1 shows that 47.2 percent of the full sample is hand-to-mouth and that the median active savings rate is zero.

Detailed interpretation: Figure 1 is the theory diagnostic: constant contribution rates are generally not optimal. Table 1 is the empirical motivation: many households have too little liquid wealth to offset constant mandated contributions. Together they explain why a rigid pension contribution schedule can be costly.

Economic message: The key policy problem is not only theoretical suboptimality; it is that a large share of households cannot undo the suboptimal rule privately.



(a) Figure 1: Optimal savings rates.

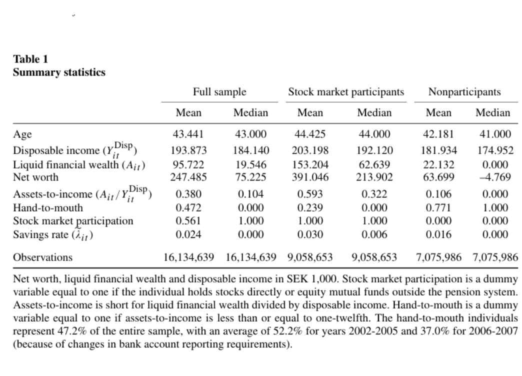


Table 1
Summary statistics

	Full sample		Stock market participants		Nonparticipants	
	Mean	Median	Mean	Median	Mean	Median
Age	43.441	43.000	44.425	44.000	42.181	41.000
Disposable income (Y_{it}^{Disp})	193.873	184.140	203.198	192.120	181.934	174.952
Liquid financial wealth (A_{it})	95.722	19.546	153.204	62.639	22.132	0.000
Net worth	247.485	75.225	391.046	213.902	63.699	–4.769
Assets-to-income ($A_{it}/Y_{it}^{\text{Disp}}$)	0.380	0.104	0.593	0.322	0.106	0.000
Hand-to-mouth	0.472	0.000	0.239	0.000	0.771	1.000
Stock market participation	0.561	1.000	1.000	1.000	0.000	0.000
Savings rate ($\tilde{\lambda}_{it}$)	0.024	0.000	0.030	0.006	0.016	0.000
Observations	16,134,639	16,134,639	9,058,653	9,058,653	7,075,986	7,075,986

Net worth, liquid financial wealth and disposable income in SEK 1,000. Stock market participation is a dummy variable equal to one if the individual holds stocks directly or equity mutual funds outside the pension system. Assets-to-income is short for liquid financial wealth divided by disposable income. Hand-to-mouth is a dummy variable equal to one if assets-to-income is less than or equal to one-twelfth. The hand-to-mouth individuals represent 47.2% of the entire sample, with an average of 52.2% for years 2002-2005 and 37.0% for 2006-2007 (because of changes in bank account reporting requirements).

2.2 The role of age and expected income growth

(b) Table 1: Summary statistics.

7.2 Figure 2 and Table 2: Expected income growth, age, and savings rates

Read:

- Figure 2 shows that savings rates decline with expected income growth and that age groups differ in slope and level.
- Table 2 confirms the negative relation between expected income growth and active savings rates.
- Non-hand-to-mouth households respond much more than hand-to-mouth households.

Detailed interpretation: This pair tests Proposition 1. Younger individuals facing high income growth should save less today. The empirical evidence broadly supports this idea, but mostly among households with sufficient liquid assets.

Economic message: The age rule in the proposed pension design is justified because voluntary liquid saving already follows an age/income-growth pattern for households that can adjust.

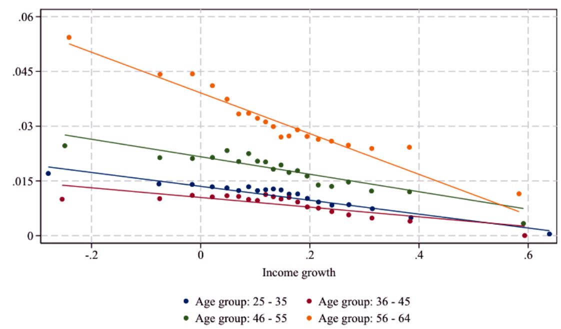


Figure 2
Income growth, age, and savings rates
The figure is a binned scatterplot of savings rates against income growth rates. The savings rate is the average

(a) Figure 2: Savings rates, age, and income growth.

Table 2
Savings rates and the role of expected income growth and age

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Full sample		Hand-to-mouth		Non-hand-to-mouth	
$y^{Disp}_{i,t+4} - y^{Disp}_{i,t}$	−0.026***	−0.056***		−0.005***		−0.037***	
	(0.001)	(0.003)		(0.000)		(0.001)	
$\mathbb{1}$[Age 25 − 35]		−0.026***	−0.025***		0.006***		−0.019***
		(0.001)	(0.000)		(0.000)		(0.000)
$\mathbb{1}$[Age 36 − 45]		−0.029***	−0.028***		0.006***		−0.025***
		(0.001)	(0.000)		(0.000)		(0.000)
$\mathbb{1}$[Age 46 − 55]		−0.017***	−0.017***		0.004***		−0.014***
		(0.001)	(0.000)		(0.000)		(0.001)
$(y^{Disp}_{i,t+4} - y^{Disp}_{i,t})\times$ $\mathbb{1}$[Age 25 − 35]		0.037***					
		(0.003)					
$(y^{Disp}_{i,t+4} - y^{Disp}_{i,t})\times$ $\mathbb{1}$[Age 36 − 45]		0.043***					
		(0.003)					
$(y^{Disp}_{i,t+4} - y^{Disp}_{i,t})\times$ $\mathbb{1}$[Age 46 − 55]		0.032***					
		(0.003)					
Constant	0.019***	0.039***	0.038***	−0.010***	−0.016***	0.048***	0.063***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
F $\mathbb{1}$[Age 25 − 35] = $\mathbb{1}$[Age 36 − 45]		143***	161***		4**		245***
F $\mathbb{1}$[Age 36 − 45] = $\mathbb{1}$[Age 46 − 55]		1,310***	2,400		317***		759***
F $\mathbb{1}$[Age 46 − 55] = $\mathbb{1}$[Age 56 − 64]		935***	2,645***		364***		721***
Observations	1,602,790	1,602,790	2,386,418	847,229	1,1907,47	755,561	1,195,671
R^2	.002	.006	.006	.000	.002	.002	.003

The table reports estimates of regressing savings rates on income growth rates and age dummy variables using OLS. The savings rate is the average over 2 years, $(\widetilde{\lambda}_{i,2002}+\widetilde{\lambda}_{i,2003})/2$ and income growth is defined as $y^{\text{Disp}}_{i,t+4} - y^{\text{Disp}}_{i,t}$, where $y^{\text{Disp}}_{i,t} = \log((Y^{\text{Disp}}_{i,2002}+Y^{\text{Disp}}_{i,2003})/2)$ and $y^{\text{Disp}}_{i,t+4} = \log((Y^{\text{Disp}}_{i,2006}+Y^{\text{Disp}}_{i,2007})/2)$. Columns 1–3 report estimates for the entire sample. Columns 4 and 5 report estimates for hand-to-mouth individuals. Columns 6 and 7 report estimates for non-hand-to-mouth individuals. * $p<.1$, ** $p<.05$, *** $p<.01$.

(b) Table 2: Expected income growth and age.

7.3 Tables 3 and 4: Balance shocks and income shocks

Read:

- Table 3 tests whether households reduce active saving after positive asset-balance shocks.
- Table 4 tests whether households raise active saving after positive income shocks.
- IV specifications address mechanical and endogeneity concerns.
- Responses are much stronger among non-hand-to-mouth households.

Detailed interpretation: These tables test Proposition 2 and the broader balance-to-income logic. A high balance relative to income should reduce the need for current contributions; a positive income shock should raise saving. The evidence supports the principle but highlights strong heterogeneity in ability to adjust.

Economic message: Balance-to-income-dependent contribution rates are valuable because many households cannot replicate them through liquid saving adjustments.

Table 3
Savings rates and the role of asset-balance shocks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS				IV		
$\frac{A_{it-1}}{Y_{it}^{Disp}} \times R_{it}^{A}$	–0.106***	–0.103***	–0.104***	–0.154***	–0.122***	–0.138***	–0.088***
	(0.006)	(0.006)	(0.006)	(0.014)	(0.008)	(0.017)	(0.008)
Age group	25-64	25-64	25-64	25-35	36-45	46-55	56-64
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F			74,704***	49,735***	32,074***	93,935***	17,881***
R^2	.318	.319					
Observations	8,900,223	8,872,650	8,872,650	2,212,862	2,282,590	2,178,616	1,820,876

This table reports estimates of regressing savings rates on an interaction between asset balance-to-income ratios and asset returns. The sample is restricted to stock market participants (i.e., individuals with nonzero stock, mutual fund or capital insurance holdings). The control variables are Y_{it}^{Disp}, NW_{it-1}, and $ND_{it,t-1}$. Standard errors, clustered at the level of the individual and the individual's largest security holding, are in parentheses. The individual's largest security holding is the particular stock or mutual fund—identified by their International Securities Identification Number (ISIN)—with the largest weight in the individual's financial asset portfolio. When the largest holding is bonds, bank accounts or capital insurance accounts, we classify the largest security holding by its respective asset type. Singleton groups are excluded. Table B.2 reports first-stage IV estimates. $^{*}p<.1$, $^{**}p<.05$, $^{***}p<.01$.

(a) Table 3: Asset-balance shocks.

Table 4
Savings rates and the role of income shocks

	(1)	(2)	(3)	(4)	(5)	(6)
	Full sample		Hand-to-mouth		Non-hand-to-mouth	
	OLS	IV	OLS	IV	OLS	IV
$y_{it}^{Disp} - y_{it-1}^{Disp}$	0.022***	0.024***	0.009***	-0.004	0.046***	0.061***
	(0.000)	(0.008)	(0.000)	(0.002)	(0.001)	(0.017)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual clusters	Yes	Yes	Yes	Yes	Yes	Yes
Employer clusters	No	Yes	No	Yes	No	Yes
First-stage F		6,950***		5,908***		3,358***
R^2	.256		.392		.296	
Observations	16,134,639	12,121,447	7,318,708	5,626,553	8,060,550	6,785,791

The table reports estimates of regressing savings rates on changes in income by OLS and IV. Columns 1 and 2 report estimates for the entire sample. Columns 3 and 4 report estimates for hand-to-mouth individuals. Columns 5 and 6 report estimates for non-hand-to-mouth individuals. The IV estimates in columns 2, 4, and 6 use the wage-bill growth of the employer as an instrument for the individual's income growth. Singleton groups are excluded. Table B.5 reports the first-stage estimates. $^{*}p<.1$, $^{**}p<.05$, $^{***}p<.01$.

(b) Table 4: Income shocks.

7.4 Tables 5 and 6 with Figures 3 and 4: Calibration and model fit

Read:

- Table 5 reports calibrated preference, income, asset-return, stock market entry, and pension parameters.
- Table 6 compares targeted data moments with model moments.
- Figures 3 and 4 show life-cycle fit for financial wealth, stock participation, consumption, income, and wealth by participation status.

Detailed interpretation: These visuals discipline the quantitative model before it is used for policy. The model matches the average asset-to-income ratio and stock participation well, and it reproduces broad life-cycle profiles. That matters because the welfare counterfactuals depend on households' liquid wealth, portfolio participation, and pension balances.

Economic message: A flexible contribution rule is only persuasive if the model can reproduce the household balance-sheet and participation patterns that determine who benefits from flexibility.

Table 5
Calibration: Model parameters

	Notation	Value
Preferences and stock market entry cost		
Discount factor[a]	β	0.941
Elasticity of intertemporal substitution	$1/\rho$	0.500
Relative risk aversion[a]	γ	14
Ceiling for stock market entry cost[a]	$\overline{\kappa}$	29,250
Floor for stock market entry cost[a]	$\underline{\kappa}$	0
Returns		
Gross risk-free rate	R_f	1.00
Equity premium	μ	0.04
Standard deviation of stock market return	σ_ε	0.18
Labor income, expense shock, and financial wealth		
Standard deviation of idiosyncratic labor income shock	σ_η	0.072
Weight of stock market shock in labor income	θ	0.040
Standard deviation of idiosyncratic expense shock	σ_ω	0.101
Standard deviation of initial labor income	σ_z	0.366
Standard deviation of initial financial wealth	σ_A	1.392
Mean of initial financial wealth	$\exp(\mu_A - \sigma_A^2/2)$	112,500
Floor for notional pension	$\underline{Y}$	85,829
Contribution rates in pension accounts		
DC account	λ	6.54%
Notional account	λ^{N}	14.95%
Life-cycle profiles		
Labor-income profile (scaled by 1.07)	g_t	[b]
Survival rates	ϕ_t	[c]

The table presents the parameter values of the model. [a]The parameter value has been determined endogenously by simulation of the model. [b]The labor-income profiles are discussed in detail in the main text. [c] The survival rates are computed from unisex statistics provided by Statistics Sweden.

(a) Table 5: Calibration parameters.

Table 6
Matched moments in data and model

	Data	Model
Financial wealth-to-labor income ratio	0.92	0.94
Stock market participation	0.51	0.49
Equity share (conditional)	0.44	0.42

The table presents matched moments in the data and model.

(b) Table 6: Matched moments.

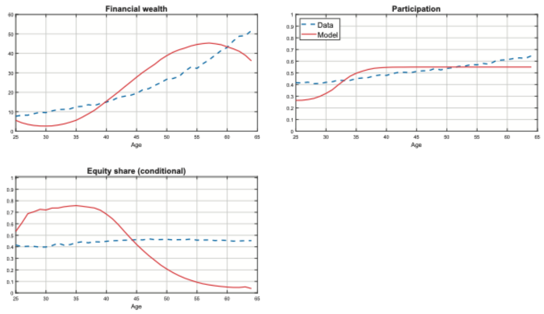


Figure 3
Calibration
The figure shows the life-cycle properties of the variables that the calibration targets (targets are their average levels). Financial wealth is expressed in SEK 10,000s.

(a) Figure 3: Calibration targets.

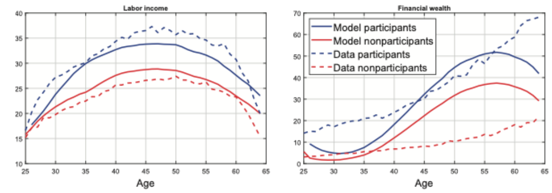


Figure 4
Model fit
The figure shows labor income and financial wealth conditional on stock market participation. Financial wealth is expressed in SEK 10,000s.

(b) Figure 4: Model fit.

7.5 Table 7 and Figure 5: Replacement rates and welfare gains

Read:

- Table 7 reports replacement rates and welfare gains for benchmark, no-DC, age-only, balance-only, and combined rules.
- The combined age and balance-to-income rule produces a mean welfare gain of about 1.8 percent.
- Figure 5 shows how welfare and replacement-rate dispersion vary across target replacement rates.
- Flexible rules reduce dispersion in replacement rates while maintaining the same average replacement rate.

Detailed interpretation: Table 7 is the main policy result. The combined rule captures both theoretical principles and therefore dominates age-only and balance-only variants. Figure 5 shows that the result is not tied to a single replacement-rate target: the flexible rules improve the welfare-dispersion tradeoff over a range of targets.

Economic message: The proposed rule improves both welfare and retirement-income risk: it avoids both under-saving and over-saving outcomes.

Table 7
Replacement rates and welfare gains

	(1) Benchmark	(2) No DC	(3) Age dependency	(4) B/Y dependency	(5) Both age and B/Y dependency
A: Replacement rate out of the DC account					
Mean	0.29	—	0.29	0.29	0.29
Standard deviation	0.12	—	0.10	0.08	0.07
95th percentile	0.54	—	0.49	0.45	0.43
5th percentile	0.15	—	0.17	0.19	0.20
B: Replacement rate out of total wealth					
Mean	0.94	0.80	0.93	0.93	0.92
Standard deviation	0.30	0.20	0.29	0.28	0.27
95th percentile	1.57	1.19	1.54	1.52	1.50
5th percentile	0.65	0.60	0.65	0.65	0.65
C: Welfare gain relative to benchmark (in percent)					
Mean	—	5.1	1.1	1.2	1.8
Standard deviation	—	0.7	0.2	0.2	0.3
95th percentile	—	6.0	1.4	1.4	2.0
5th percentile	—	3.9	0.7	0.7	1.3

Panel A reports moments of replacement rates out of the DC account ($h^B(B_{65}/Y_{64})$. Panel B reports moments of replacement rates out of total wealth ($(h^B(B_{65}+A_{65})+h^N(N_{65}))/Y_{64}$). Panel C reports moments of ex ante welfare gains associated with a shift from the benchmark to each one of the other DC plan designs. The age-dependent policy in column 3 corresponds to $\lambda_{it}=0.0104+0.003t$. The B/Y-dependent policy in column 4 corresponds to $\lambda_{it}=0.0645-0.2\left(\frac{B_{it}}{Y_{it}}/\chi_t-1\right)$. The policy in column 5 corresponds to $\lambda_{it}=0.0101+0.003t-$

(a) Table 7: Replacement rates and welfare gains.

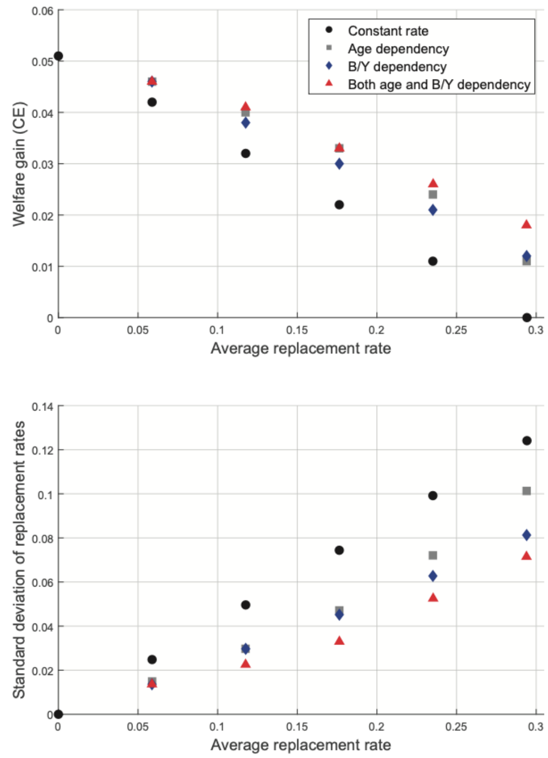


Figure 5
Replacement rate targets, welfare gains, and dispersion of replacement rates
The figure shows welfare gains (consumption equivalents) and the dispersion of replacement rates out of the DC account against target replacement rates for four types of contribution rate policies: (1) constant contribution rates ($\lambda_{it}=\beta_0$), (2) only age coefficients ($\lambda_{it}=\beta_0+\beta_1 t$), (3) only B/Y coefficients ($\lambda_{it}=\beta_0+\beta_2(\frac{B_{it}}{Y_{it}}/\chi_t-1)$, and (4) age and B/Y coefficients ($\lambda_{it}=\beta_0+\beta_1 t+\beta_2(\frac{B_{it}}{Y_{it}}/\chi_t-1)$).

(b) Figure 5: Welfare and replacement-rate dispersion.

7.6 Figure 6 and Table 8: How the optimal design changes behavior

Read:

- Figure 6 shows contribution rates and DC account balances under the benchmark and flexible pension system.
- Flexible contributions are lower early in life and increase later, with additional adjustment to the balance-to-income ratio.
- Table 8 shows effects on consumption, financial wealth, stock market participation, and DC balances at ages 30, 40, 50, and 60.

Detailed interpretation: Figure 6 shows the mechanism of the welfare gains: contribution rates are shifted away from young constrained ages and toward later ages or low-balance states. Table 8 shows the behavioral consequences: more liquid resources early in life, changed participation incentives, and more targeted DC accumulation.

Economic message: The flexible rule gives cash-flow relief when it is most valuable and forces catch-up saving when the DC balance falls behind.

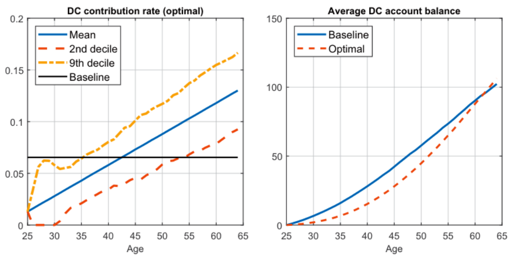


Figure 6
Benchmark versus flexible pension system
The left panel shows the average and the 2nd and 9th deciles of the contribution rate into the DC account (i.e., λ_{it}) for the optimal age- and balance-to-income-dependent design compared to the flat contribution rates of the benchmark design. The right panel shows the average DC account balance during working life. Values are expressed in SEK 10,000s.

(a) Figure 6: Benchmark versus flexible system.

Table 8
Effect of the optimal design on behavior

	Age 30	Age 40	Age 50	Age 60
A: Average Consumption				
Benchmark	20.6	26.4	27.7	25.0
Optimal	21.4	26.5	27.1	24.3
Changes (percent)	3.9	0.4	−2.2	−2.8
B: Average Financial Wealth				
Benchmark	2.7	15.4	38.8	43.4
Optimal	2.9	16.7	38.2	37.2
Changes (percent)	7.4	8.4	−1.5	−14.3
C: Average Stock Market Participation				
Benchmark	0.32	0.55	0.55	0.55
Optimal	0.29	0.39	0.39	0.39
Changes (percent)	−9.4	−29.1	−29.1	−29.1

The table reports average consumption, financial wealth, and stock market participation at different points in the life cycle. It compares the levels in the benchmark calibration and under the optimal design of contribution rates (in SEK 10,000s) and computes the change from baseline to optimal in percent.

(b) Table 8: Behavior under the optimal design.

7.7 Table 9 and Figures 7-8: Shocks, temptation, and preference robustness

Read:

- Table 9 shows how flexible contributions change the standard deviation of disposable-income changes and their correlation with labor-income and stock-market shocks.
- Figure 7 shows that welfare gains remain positive under temptation/self-control preferences.
- Figure 8 shows that welfare gains remain positive across risk aversion and EIS values.

Detailed interpretation: Flexible contributions mechanically link disposable income to account balances and returns, so Table 9 is needed to document new cash-flow risk. Figures 7 and 8 then show that the welfare benefits survive behavioral and preference misspecification concerns. This is important because mandatory pensions are often justified precisely by self-control problems.

Economic message: The proposed rule is not fragile: it remains beneficial even when households face temptation and when preference parameters vary widely.

Table 9
Changes in disposable income: Standard deviation and correlation with shocks

	Standard deviation	Correlation with changes in gross income	Correlation with stock Market returns
Benchmark	21.24	1.00	−0.0014
Optimal	19.26	0.95	−0.0074

The table reports statistics of changes in disposable income: average lifetime standard deviation, correlation with changes in gross income, correlation with changes in stock market returns.

(a) Table 9: Disposable-income shocks.

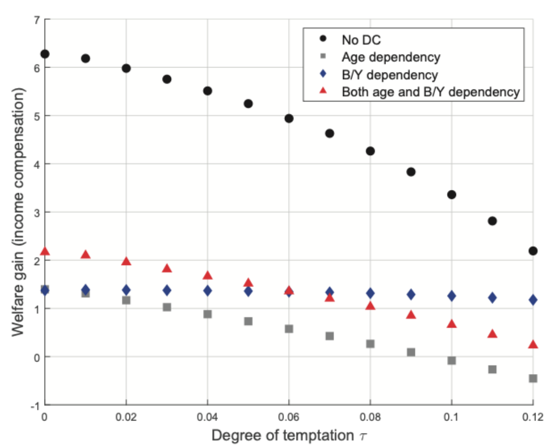


Figure 7
Temptation and welfare gains
The figure shows the welfare gains (expressed as income compensation) of the alternative designs of contribution rates for different degrees of temptation. Note that, for each degree of temptation, the model has been recalibrated to match data moments (hence, preference parameters β and $\bar{\kappa}$ vary across different horizons).

(b) Figure 7: Temptation and welfare gains.

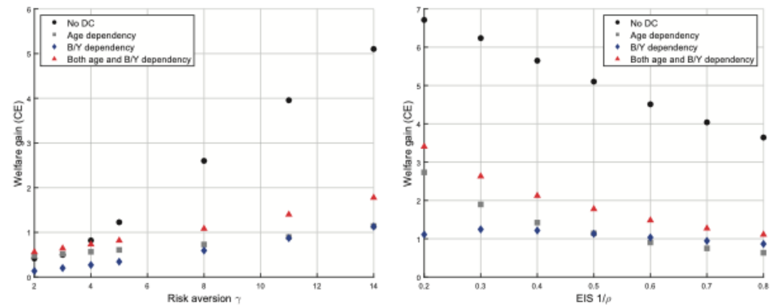


Figure 8
Preference parameters and welfare gains
The figure shows the welfare gains (expressed as consumption equivalent) of the alternative designs of contribution rates for different degrees levels of risk aversion (left panel) and elasticity of intertemporal substitution (right panel). Note that, for each specified parameter value, the model has been recalibrated to match data moments (hence, preference parameters β and $\bar{\kappa}$ vary across different horizons).

Figure 8: Preference parameters and welfare gains.

7.8 Table 10: Alternative policy margins

Read:

- Table 10 compares contribution-rule redesign to changing mandatory equity shares or annuitization fractions.
- The combined contribution rule yields a mean welfare gain of 1.8 percent.
- Changing the investment strategy or annuitization margin yields effects close to zero by comparison.

Detailed interpretation: Table 10 is the paper's policy-priority table. It shows that contribution-rate flexibility is not just one small improvement among many. In this calibrated economy, it is more than an order of magnitude more important than changing equity-share or annuitization rules.

Economic message: Policymakers should prioritize contribution-rate design, not only asset allocation or decumulation design.

Table 10
Welfare gains of changing alternative policy margins

	Both age and B/Y dependency	Mandatory equity share		Annuitization fraction	
		Constant	DSV 2018	No annuities	50% annuitized
	(1)	(2)	(3)	(4)	(5)
Mean	1.8	−0.02	0.05	−0.01	0.04
Standard deviation	0.3	0.01	0.04	0.04	0.01
95th percentile	2.0	−0.01	0.07	0.03	0.05
5th percentile	1.3	−0.04	0.02	−0.08	0.03

The table reports moments of welfare gains (in percent of consumption equivalent) for policies that change alternative margins of mandatory DC pension plans. For comparison, column 1 repeats the welfare effects of our preferred rule for contribution rates (the combined rule). Columns 2 and 3 show the welfare effects changing the investment: a constant mandatory equity share throughout working life (column 2) and the optimal mandatory equity share according to Dahlquist, Setty, and Vestman (2018) (column 3). Columns 4 and 5 vary the fraction that of the DC balance that is annuitized at entry to retirement: no annuitization (column 4) and 50% annuitization (column 5). The comparison in all cases is relative to the benchmark economy with constant contribution rates, a "100-minus-age" equity share, and full amortization during retirement.

Table 10: Welfare gains of changing alternative policy margins.

8 Result map

- **Theory result 1.** Young-age savings rates should decline with expected income growth.
- **Theory result 2.** Middle-age savings rates should decline with the balance-to-income ratio.
- **Empirical result 1.** Swedish liquid savings behavior is broadly consistent with the theory for households that are not hand-to-mouth.
- **Empirical result 2.** Almost half of individuals are hand-to-mouth and cannot undo rigid mandatory constant contributions.
- **Model result 1.** A flexible contribution rule based on age and balance-to-income yields about 1.8 percent welfare gain.
- **Model result 2.** The rule lowers replacement-rate dispersion while maintaining average replacement rates.
- **Robustness result.** Gains survive temptation preferences and a wide range of risk-aversion and EIS parameters.
- **Policy result.** Contribution-rate redesign is much more important than changing the mandatory investment strategy or annuitization margin.

9 Short summary

- Constant DC contribution rates conflict with consumption-savings theory.
- Optimal contribution rates should increase with age and fall with the balance-to-income ratio.
- Swedish registry data show that many households follow these principles in liquid saving, but hand-to-mouth households cannot.
- A quantitative life-cycle model with a detailed Swedish pension system evaluates policy counterfactuals.
- A simple age and balance-to-income contribution rule improves welfare by about 1.8 percent and lowers replacement-rate dispersion.

- The result is robust to temptation/self-control, preference parameters, and alternative pension-design margins.

10 Conclusion

10.1 Core conclusion

Schlafmann, Setty, and Vestman show that contribution rates in mandatory DC pension plans should not be constant. They should rise with age and fall with the DC balance-to-income ratio. A simple rule based on these two variables yields large welfare gains and reduces replacement-rate dispersion.

10.2 Economic intuition

Young workers with rising income should not be forced to save too much too early. Workers who already have high pension balances relative to income should be allowed to contribute less, while workers who fall behind should contribute more. This is just consumption smoothing applied to pension-plan design.

10.3 Incremental contribution

The paper contributes by translating consumption-savings theory into a concrete pension contribution rule, showing empirically who can and cannot implement the rule privately, and using a quantitative model to show that the welfare gains exceed those from changing investment or annuitization policy.

10.4 Policy implications

DC pension reforms should focus on contribution-rate design. Auto-escalation and defaults should be conditional on age and account balances, not only constant percentages of wages. Such rules are implementable because pension administrators observe age, income, and account balances.

10.5 Limitations

The welfare results rely on a calibrated model and on the Swedish institutional setting. The model does not capture all possible labor-supply responses, housing decisions, or political constraints. However, the simplicity of the proposed rule makes it adaptable to other settings.

10.6 Takeaway

The main takeaway is that pension plans should use the information they already have. Age and account balances are enough to make mandatory contributions much closer to optimal consumption-savings behavior.